```
%pip install matplotlib pandas numpy seaborn
```

```
Requirement already satisfied: matplotlib in .\AppData\Local\Programs\
Python\Python314\Lib\site-packages (3.11.2)
Requirement already satisfied: pandas in .\AppData\Local\Programs\
Python\Python314\Lib\site-packages (3.1.0)
Requirement already satisfied: numpy in .\AppData\Local\Programs\
Python\Python314\Lib\site-packages (2.5.4)
Requirement already satisfied: seaborn in .\AppData\Local\Programs\
Python\Python314\Lib\site-packages (0.13.2)
Requirement already satisfied: contourpy>=1.0.1 in .\AppData\Local\
Programs\Python\Python314\Lib\site-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in .\AppData\Local\
Programs\Python\Python314\Lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in .\AppData\Local\
Programs\Python\Python314\Lib\site-packages (from matplotlib) (4.68.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\AppData\Local\
Programs\Python\Python314\Lib\site-packages (from matplotlib) (1.5.1)
Requirement already satisfied: packaging>=20.0 in .\AppData\Local\
Programs\Python\Python314\Lib\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in .\AppData\Local\Programs\
Python\Python314\Lib\site-packages (from matplotlib) (13.0.0)
Requirement already satisfied: pyparsing>=3 in .\AppData\Local\
Programs\Python\Python314\Lib\site-packages (from matplotlib) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in .\AppData\
Local\Programs\Python\Python314\Lib\site-packages (from matplotlib)
(4.0.0)
Requirement already satisfied: tzdata in .\AppData\Local\Programs\
Python\Python314\Lib\site-packages (from pandas) (2026.6)
Note: you may need to restart the kernel to use updated packages.
```

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import seaborn as sns
%matplotlib inline
```

```
cartoon=pd.read_csv("C:/Users/saara/Downloads/archive (1).zip")
cartoon
```

```
                                          Name          Span  \
0                    Batman: The Animated Series  (1992–1995)
1                                  Samurai Jack  (2001–2017)
2                     Avatar: The Last Airbender  (2005–2008)
3                              Total Drama Island  (2007–2014)
4               Scooby-Doo! Mystery Incorporated  (2010–2013)
..                                           ...           ...
95                                 Viking Skool       (2022)
96                         The Looney Tunes Show  (2011–2015)
97                         The Fairly OddParents  (2001–2017)
```

```
98  The Adventures of Jimmy Neutron, Boy Genius  (2002–2006)
99                                        Daria  (1997–2002)

                                          Description  Rating
0   The Dark Knight battles crime in Gotham City w...     9.0
1   A samurai, sent through time, fights to return...     8.5
2   In a war-torn world of elemental magic, a youn...     9.3
3   Animated satire of survivor reality shows feat...     7.4
4   This incarnation finds Scooby and the gang liv...     8.1
..                                                 ...     ...
95  A group of young students in training to becom...     6.4
96  An updated iteration of the classic Looney Tun...     7.7
97  After being tortured and humiliated by his bab...     7.2
98  A young boy, who happens to be a genius, lives...     6.7
99  A smart and cynical girl goes through teenage ...     8.1

[100 rows x 4 columns]
```

```
cartoon.head()
```

```
                              Name         Span  \
0       Batman: The Animated Series  (1992–1995)
1                     Samurai Jack  (2001–2017)
2        Avatar: The Last Airbender  (2005–2008)
3                Total Drama Island  (2007–2014)
4  Scooby-Doo! Mystery Incorporated  (2010–2013)

                                          Description  Rating
0  The Dark Knight battles crime in Gotham City w...     9.0
1  A samurai, sent through time, fights to return...     8.5
2  In a war-torn world of elemental magic, a youn...     9.3
3  Animated satire of survivor reality shows feat...     7.4
4  This incarnation finds Scooby and the gang liv...     8.1
```

```
cartoon.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 4 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Name         100 non-null    str
 1   Span         100 non-null    str
 2   Description  100 non-null    str
 3   Rating       100 non-null    float64
dtypes: float64(1), str(3)
memory usage: 3.3 KB
```

##HISTOGRAM##

```
import matplotlib.pyplot as plt
import pandas as pd

# Select numerical columns
numeric_cols = cartoon.select_dtypes(include='number')

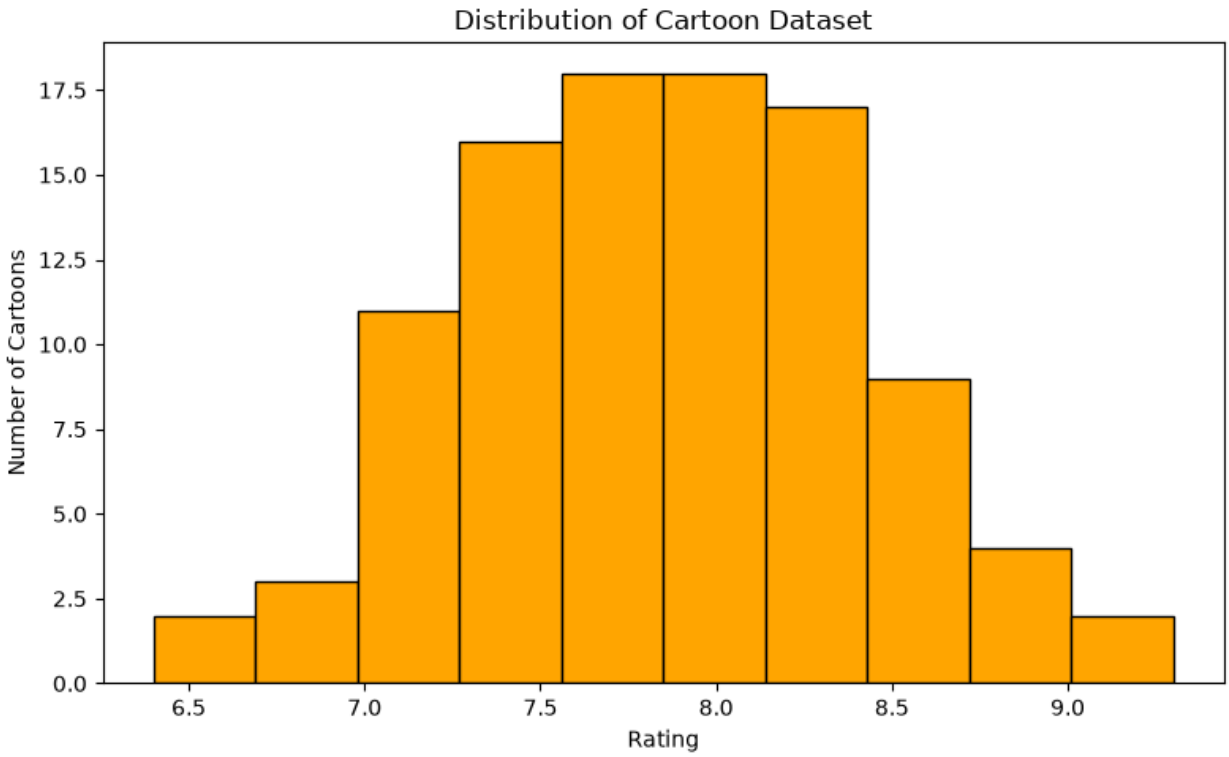
# Check numerical columns
if numeric_cols.empty:
    print("No numerical columns found!")
else:
    col = numeric_cols.columns[0]

    plt.figure(figsize=(8,5))

    plt.hist(
        cartoon[col].dropna(),
        bins=10,
        color="orange",
        edgecolor="black"
    )

    plt.title("Distribution of Cartoon Dataset")
    plt.xlabel(col)
    plt.ylabel("Number of Cartoons")

    plt.tight_layout()
    plt.show()
```

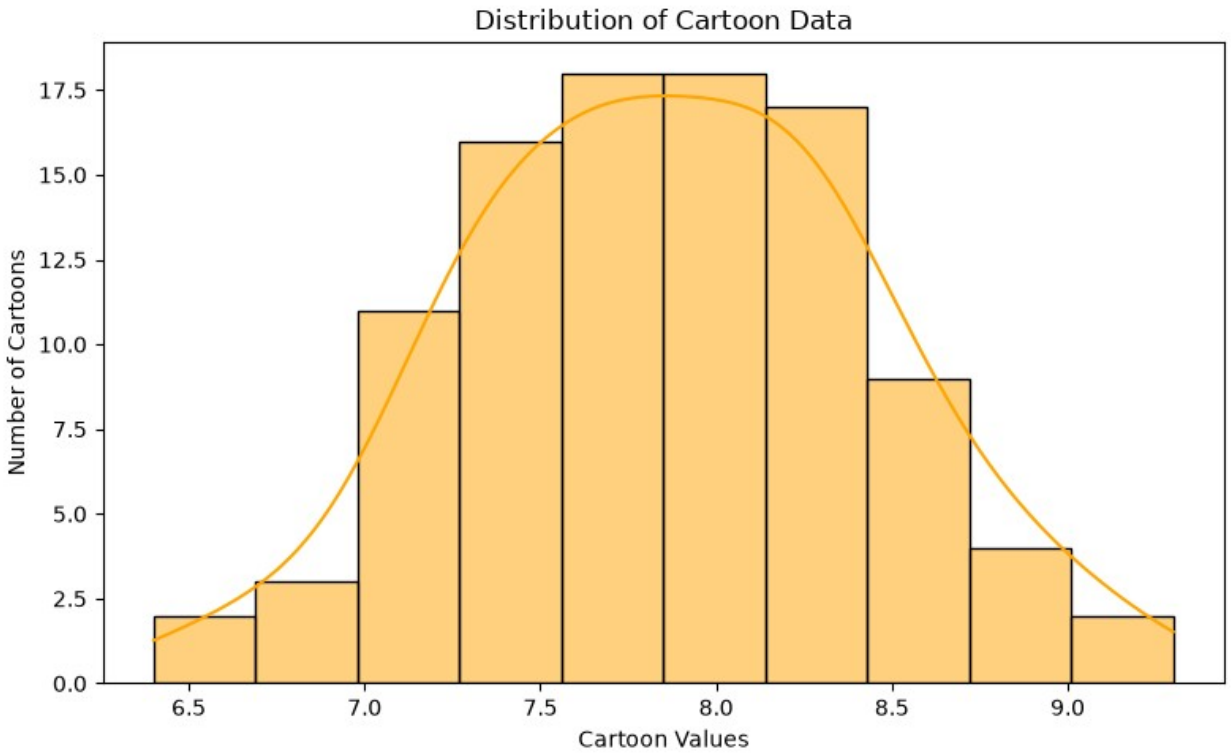



```
plt.figure(figsize=(8,5))

sns.histplot(
    cartoon.select_dtypes(include='number').iloc[:, 0].dropna(),
    bins=10,
    color="orange",
    edgecolor="black",
    kde=True
)

plt.title("Distribution of Cartoon Data")
plt.xlabel("Cartoon Values")
plt.ylabel("Number of Cartoons")

plt.tight_layout()
plt.show()
```

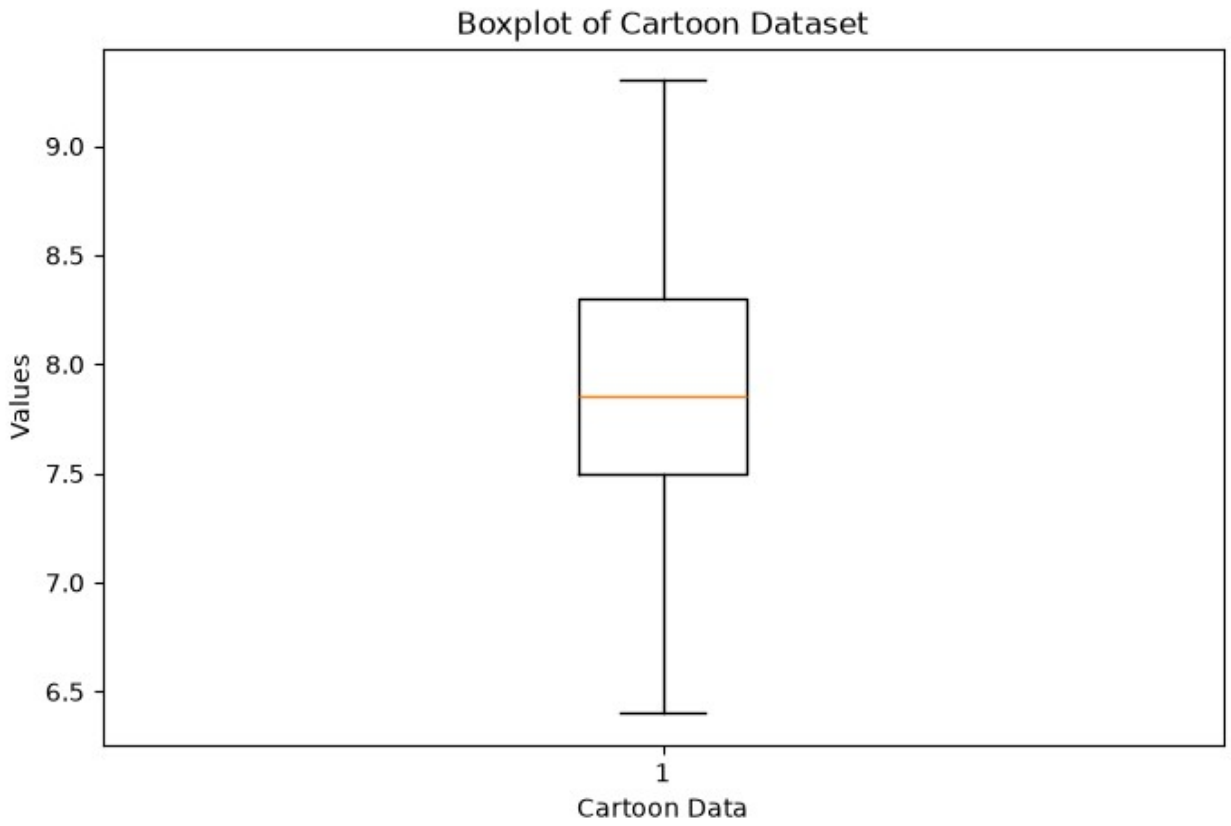



```
plt.figure(figsize=(8,5))

plt.boxplot(
    cartoon.select_dtypes(include='number').iloc[:, 0].dropna()
)

plt.title("Boxplot of Cartoon Dataset")
plt.xlabel("Cartoon Data")
plt.ylabel("Values")

plt.show()
```




```
import matplotlib.pyplot as plt

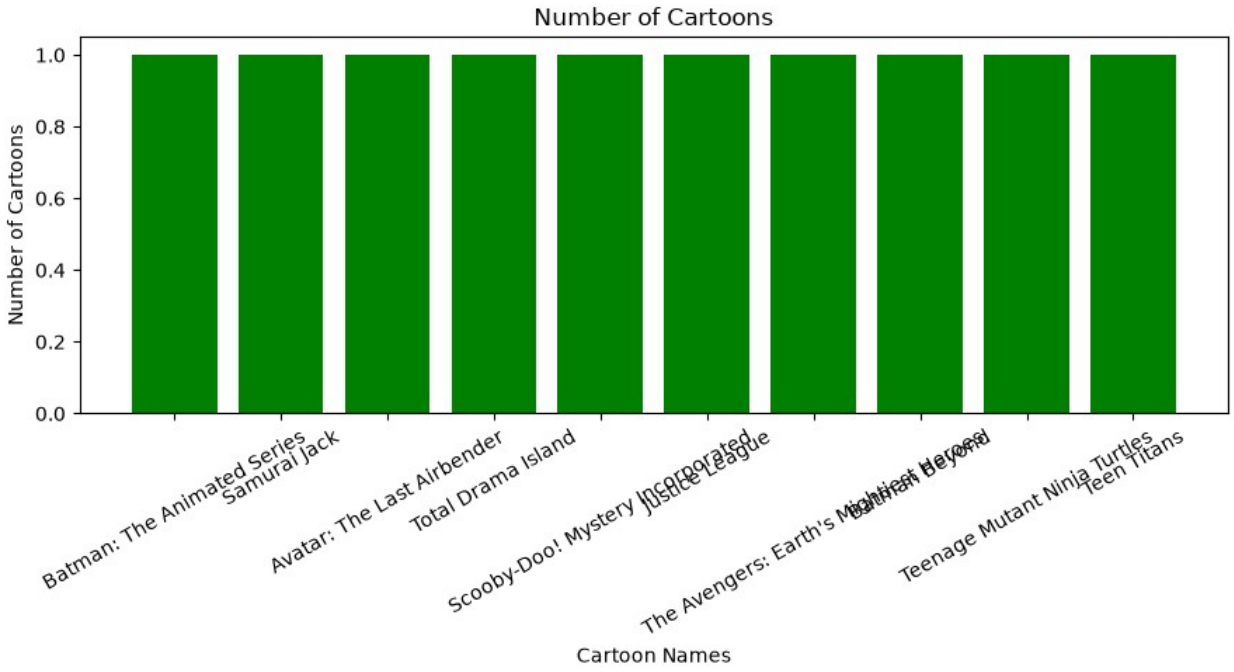
cartoon_counts = cartoon.iloc[:, 0].value_counts().head(10)

plt.figure(figsize=(9,5))

plt.bar(
    cartoon_counts.index,
    cartoon_counts.values,
    color="green"
)

plt.title("Number of Cartoons")
plt.xlabel("Cartoon Names")
plt.ylabel("Number of Cartoons")

plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```




```
import matplotlib.pyplot as plt

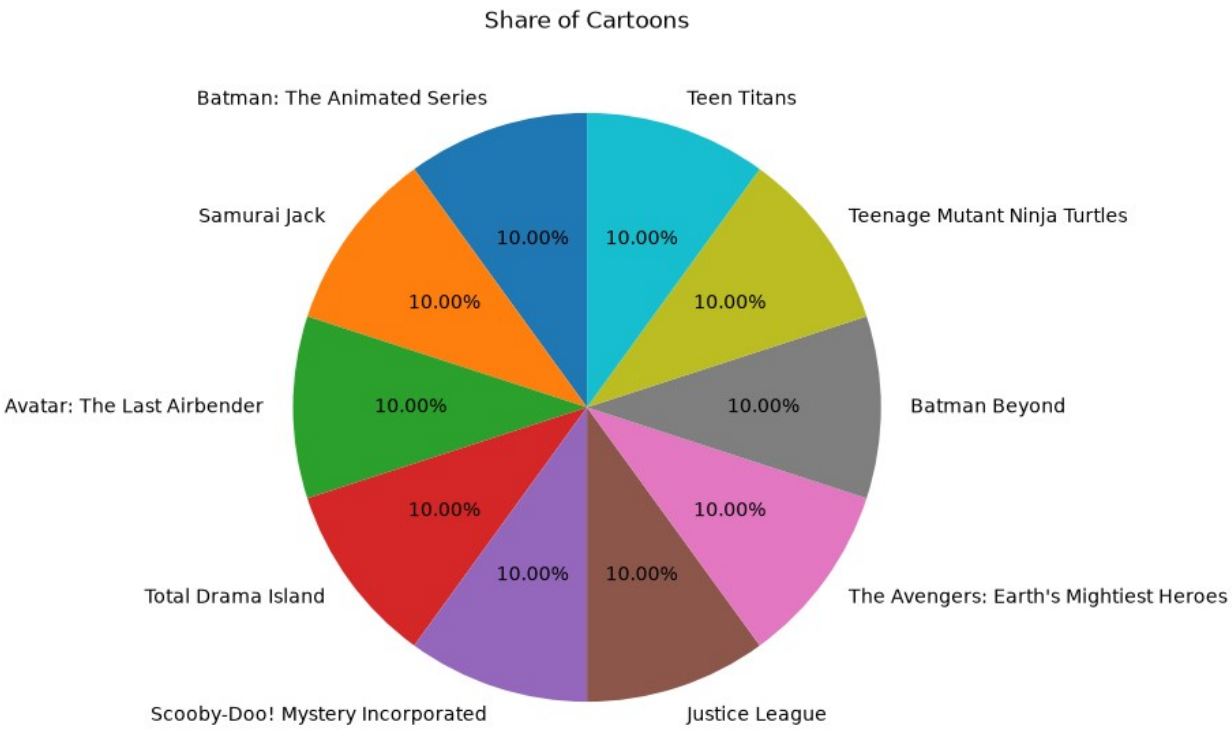
cartoon_counts = cartoon.iloc[:, 0].value_counts().head(10)

plt.figure(figsize=(7,7))

plt.pie(
    cartoon_counts,
    labels=cartoon_counts.index,
    autopct="%1.2f%%",
    startangle=90
)

plt.title("Share of Cartoons")

plt.show()
```


Share of Cartoons
Batman: The Animated Series
Teen Titans
Samurai Jack
Teenage Mutant Ninja Turtles
Avatar: The Last Airbender
Batman Beyond
Total Drama Island
The Avengers: Earth's Mightiest Heroes
Scooby-Doo! Mystery Incorporated
Justice League
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